

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
- (b) Design a responsive navigation bar using appropriate CSS techniques.
- (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
- (d) Evaluate the usability and suggest improvements for better user experience.

Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
- (b) Suggest appropriate **CSS animation or transition** for product interaction.
- (c) Design a hover effect for a product card.
- (d) Evaluate performance and user experience considerations.

Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).

- (b) Compare **Flexbox and Grid** based on layout needs.
- (c) Choose the most suitable layout technique and justify your choice.
- (d) Evaluate the scalability and responsiveness of the chosen approach.

Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
- (b) Design a mobile-first layout using CSS.
- (c) Demonstrate how the layout scales for larger screens using media queries.
- (d) Evaluate accessibility and readability of the design.

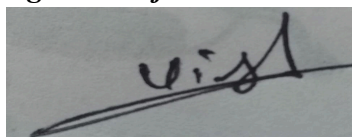
Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

Code of Conduct:

I (B.Vishnu Vardhan/ 192311267) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design requirements and concepts	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation

Question 1: Responsive Navigation Bar Design (4 Marks)

(a) Identify Key Components

A responsive navigation bar for an e-commerce website should include:

- Website Logo
- Home
- Products/Categories
- Search Bar
- Shopping Cart
- Login/Register
- Contact Us
- Hamburger Menu for mobile devices

(b) Design

Use **CSS Flexbox** to align navigation items horizontally and **media queries** to make the layout responsive.

Sample HTML

```
<nav class="navbar">
  <h2>ShopEasy</h2>

  <div class="menu">
    <a href="#">Home</a>
    <a href="#">Products</a>
    <a href="#">Cart</a>
    <a href="#">Contact</a>
  </div>
</nav>
```

Sample CSS

```
.navbar{
display:flex;
justify-content:space-between;
align-items:center;
background:#333;
padding:15px;
}

.menu{
display:flex;
gap:20px;
}
```

(c) Demonstration

```
@media(max-width:768px){  
  
  .menu{  
    display:none;  
    flex-direction:column;  
  }  
  
  .menu.show{  
    display:flex;  
  }  
  
}
```

Result:

- Desktop → Horizontal navigation
- Tablet → Flexible navigation
- Mobile → Hamburger menu

(d) Evaluation

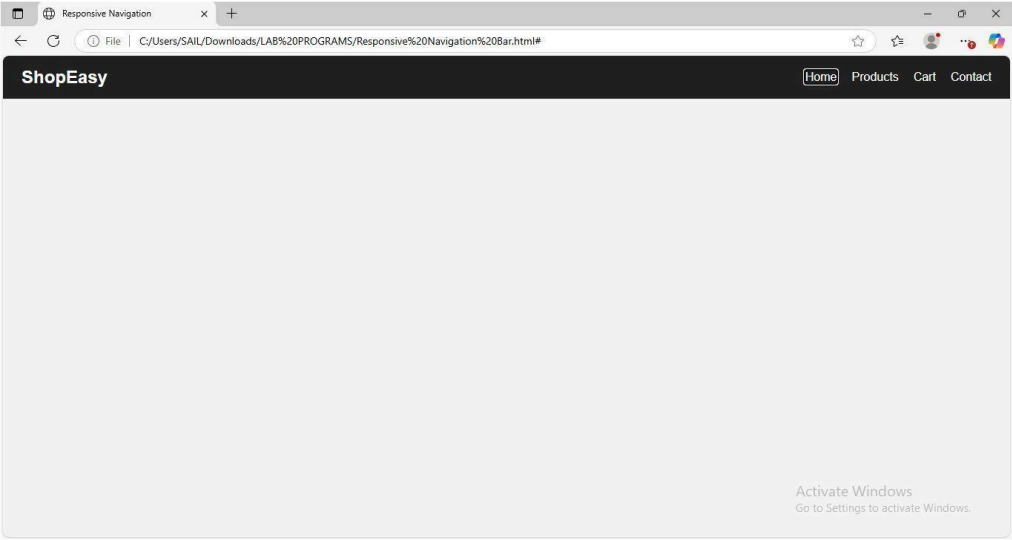
Advantages

- Mobile-friendly
- Easy navigation
- Better user experience
- Responsive across all devices

Suggested Improvements

- Animated menu
- Sticky navigation
- Search suggestions
- Dark mode

Output:



Question 2: Product Hover Effect (4 Marks)

(a) CSS Properties

Suitable properties include:

- transform
- transition
- box-shadow
- opacity
- border-radius
- background-color

(b) Animation

Use CSS Transition with Scale Effect.

```
transition:0.3s;  
transform:scale(1.05);
```

(c) Design

```
.card{  
  
width:220px;  
  
padding:20px;  
  
box-shadow:0 0 10px gray;  
  
transition:0.4s;  
  
}  
  
.card:hover{  
  
transform:scale(1.05);  
  
box-shadow:0 0 20px black;  
  
}
```

(d) Evaluation

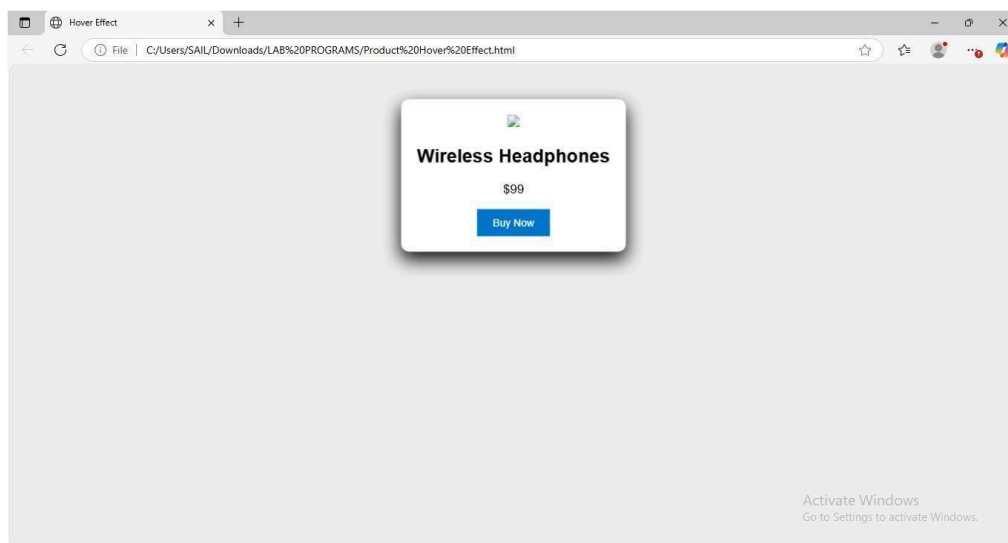
Advantages

- Better product visibility
- Interactive interface
- Attractive appearance

Improvements

- Use subtle animations
- Optimize for low-end devices
- Avoid excessive visual effects

Output:



Question 3: Layout Selection (Flexbox vs Grid)

(a) Dashboard Requirements

A dashboard generally contains:

- Header
- Sidebar
- Main Content
- Cards
- Footer

It requires:

- Rows
- Columns
- Proper alignment
- Equal spacing

(b) Comparison

Flexbox	Grid
One-dimensional	Two-dimensional
Best for rows or columns	Best for complete page layouts
Easier alignment	Better control over layout
Suitable for navigation	Suitable for dashboards

(c) Selecti

on CSS Grid

Reason:

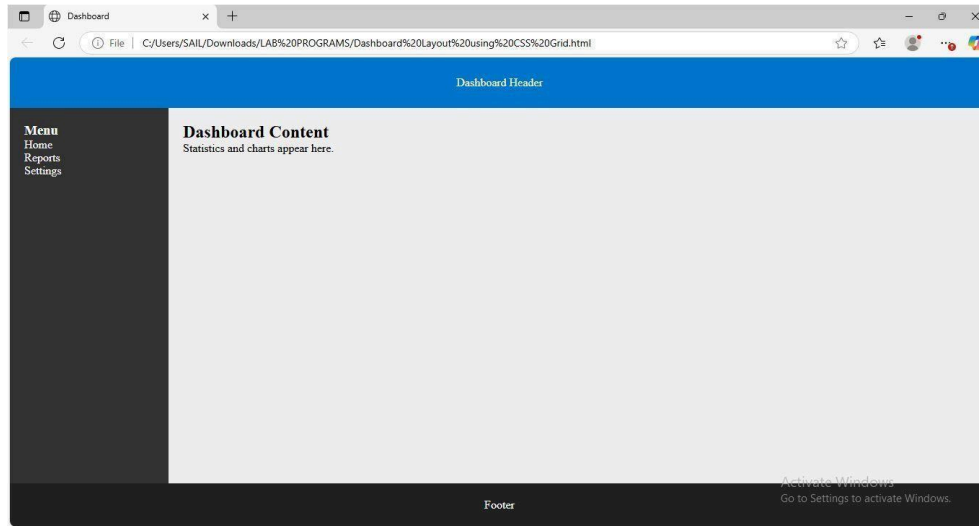
- Supports both rows and columns
- Better dashboard organization
- Easier responsive design

(d) Evaluation

Advantages

- Scalable
- Responsive
- Easy maintenance
- Cleaner code

Output:



Question 4: Mobile-First Blog Layout

(a) Key Sections

- Header
- Navigation
- Blog Content
- Sidebar
- Footer

(b) Design

```
.container{  
  
display:flex;  
  
flex-direction:column;  
  
}
```

Mobile devices display a single-column layout.

(c) Media Query

```
@media(min-width:768px){  
  
.container{  
  
flex-direction:row;  
  
}  
  
}
```

Desktop displays two columns.

(d) Evaluation

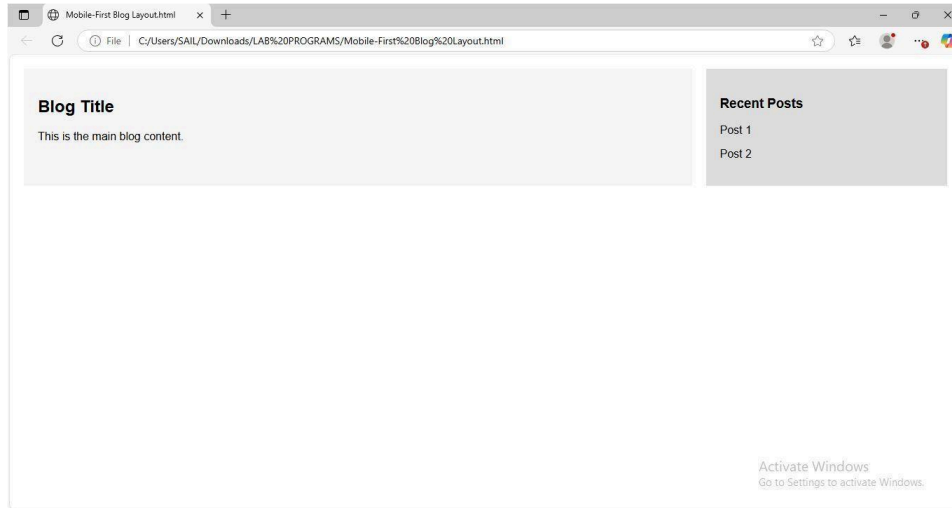
Advantages

- Better readability
- Mobile-friendly
- Faster loading
- Accessible

Improvements

- Dark mode
- Larger fonts
- Better spacing
- Responsive images

Output:



Question 5: JavaScript Event Handling

(a) Events

Suitable events include:

- click
- mouseover
- mouseout
- keydown

(b) Event Logic

Algorithm:

1. User clicks menu icon.
2. JavaScript checks menu status.
3. If hidden, display menu.
4. Otherwise, hide menu.

(c) Implementation

```
function toggleMenu(){  
  
let menu=document.getElementById("menu");  
  
menu.classList.toggle("show");  
  
}
```

(d) Evaluation

Advantages

- Fast response
- Easy implementation
- Improved navigation
- Better mobile usability

Suggested Improvements

- Keyboard accessibility
- Smooth animations

- ARIA labels
- Touch gesture support

Output:

